

FCFS Scheduling

```
shamruth@DESKTOP-HHJO  ×  +  ▾
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort processes according to burst time
    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (bt[i] > bt[j])
            {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");
}
```

Output

```
shamruth@DESKTOP-HHJODKP:~$ nano fcfs.c
shamruth@DESKTOP-HHJODKP:~$ gcc fcfs.c -o fcfs
./fcfs
Enter number of processes: 7
Enter burst time:
P1: 4
P2: 5
P3: 7
P4: 7
P5: 5
P6: 8
P7: 9

Process BT      WT      TAT
P1      4      0      4
P2      5      4      9
P3      7      9     16
P4      7     16     23
P5      5     23     28
P6      8     28     36
P7      9     36     45
shamruth@DESKTOP-HHJODKP:~$ |
```

SJF Non-Preemptive

```
shammruth@DESKTOP-HHJO  ×  +  ▾
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort processes according to burst time
    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (bt[i] > bt[j])
            {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++)
    {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++)
    {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    ^G Help      ^O Write Out   ^F Where Is
    ^X Exit      ^R Read File   ^\ Replace
```

Out put

```
shamruth@DESKTOP-HHJODKP:~$ nano sjf.c
shamruth@DESKTOP-HHJODKP:~$ gcc sjf.c -o sjf
./sjf
Enter number of processes: 2
Enter burst time:
P1: 3
P2: 5

Process BT      WT      TAT
P1      3      0      3
P2      5      3      8
shamruth@DESKTOP-HHJODKP:~$ |
```

Priority Scheduling

```
shammruth@DESKTOP-HHJO  ×  +  v
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i, j;
    int bt[20], priority[20];
    int wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

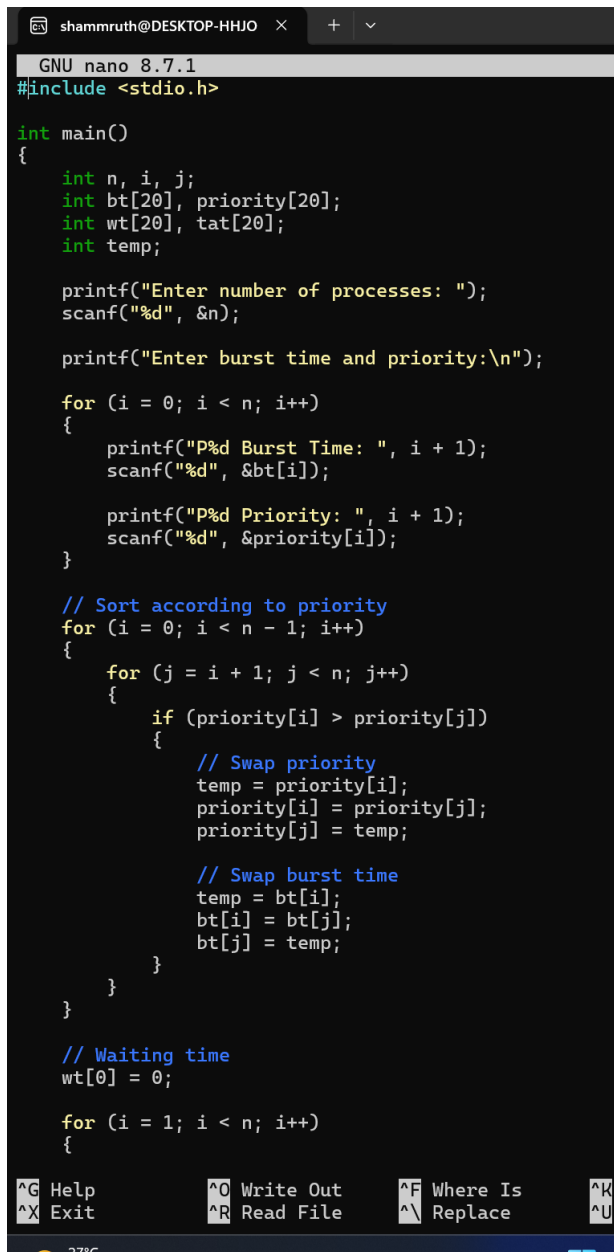
    // Sort according to priority
    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (priority[i] > priority[j])
            {
                // Swap priority
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                // Swap burst time
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++)
    {

```



Out put

```
shamruth@DESKTOP-HHJODKP:~$ nano priority.c
shamruth@DESKTOP-HHJODKP:~$ gcc priority.c -o priority
./priority
Enter number of processes: 4
Enter burst time and priority:
P1 Burst Time: 4
P1 Priority: 4
P2 Burst Time: 4
P2 Priority: 4
P3 Burst Time: 5
P3 Priority: 6
P4 Burst Time: 7
P4 Priority: 4
```

Process	BT	Priority	WT	TAT
P1	4	4	0	4
P2	4	4	4	8
P3	7	4	8	15
P4	5	6	15	20

```
shamruth@DESKTOP-HHJODKP:~$ |
```

Round Robin

```
shammruth@DESKTOP-HHJO  ×  +  ▾
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);

        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    // Round Robin Scheduling
    while (1)
    {
        done = 1;

        for (i = 0; i < n; i++)
        {
            if (rem_bt[i] > 0)
            {
                done = 0;

                if (rem_bt[i] > quantum)
                {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
                else
                {
                    time = time + rem_bt[i];
                }
            }
        }
    }
}
```

[^]G Help [^]O Write Out [^]F Where Is
[^]X Exit [^]R Read File [^]\ Replace

Out put

```
shammruth@DESKTOP-HHJODKP:~$ nano roundrobin.c
shammruth@DESKTOP-HHJODKP:~$ gcc roundrobin.c -o roundrobin
./roundrobin
Enter number of processes: 4
Enter burst time:
P1: 4
P2: 4
P3: 4
P4: 4
Enter time quantum: 4

Process BT      WT      TAT
P1      4       0       4
P2      4       4       8
P3      4       8      12
P4      4      12      16
shammruth@DESKTOP-HHJODKP:~$ |
```